THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · EASY LEVEL UNIT TEST SET 1

Chapters 11–14: **Exploring Some Geometric Themes, Tales by Dots and Lines, Algebra Play, Area** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. In the Sierpinski triangle construction, the number of small shaded triangles at stage 1 (stage 0 has one) is: 1

(a) 3 **(b)** 6 **(c)** 4 **(d)** 2

2. The mean of 30, 25, 17, 9, 24 is: 1

(a) 19 **(b)** 42 **(c)** 23 **(d)** 21

3. The expression for '5 more than 2 times a number x' is: 1

(a) $2 + 7x$ **(b)** $5x + 2$ **(c)** $2x + 5$ **(d)** $2(x + 5)$

4. The area of a rectangle 12 cm long and 7 cm broad is: 1

(a) 84 cm² **(b)** 96 cm² **(c)** 19 cm² **(d)** 38 cm²

5. The number of faces of a triangular prism is: 1

(a) 4 **(b)** 3 **(c)** 5 **(d)** 6

6. The median of 28, 19, 30, 34, 21, 38, 8 is: 1

(a) 25.43 **(b)** 8 **(c)** 28 **(d)** 34

7. The value of $3x + 4$ when $x = 5$ is: 1

(a) 19 **(b)** 17 **(c)** 38 **(d)** 21

8. The area of a triangle with base 10 cm and height 6 cm is: 1

(a) 60 cm² **(b)** 16 cm² **(c)** 30 cm² **(d)** 40 cm²

9. The number of vertices of a triangular pyramid is: 1

(a) 5 **(b)** 6 **(c)** 3 **(d)** 4

10. The mode of 18, 16, 20, 18, 18, 9, 8 is: 1

(a) 8 **(b)** 18 **(c)** 20 **(d)** 16

11. On simplifying $2x + 3 + 4x + 5$ we get: 1

(a) $8x + 8$ **(b)** $6x + 8$ **(c)** $14x$ **(d)** $6x + 15$

12. The area of a parallelogram with base 9 cm and height 4 cm is: 1

(a) 18 cm² **(b)** 13 cm² **(c)** 26 cm² **(d)** 36 cm²

13. The number of edges of a triangular prism is: 1

(a) 8 **(b)** 5 **(c)** 6 **(d)** 9

14. The range of 36, 30, 20, 53, 50, 45 is: 1

(a) 33 **(b)** 53 **(c)** 73 **(d)** 39

15. The two-digit number with tens digit x and ones digit 2 is written as: 1

(a) $10x + 2$ **(b)** $20 + x$ **(c)** $x + 2$ **(d)** $2x$

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Write the number of faces, vertices and edges of a triangular prism. 2

17. Find the mean of 38, 45, 19, 18, 20. 2

18. Write an algebraic expression for: the sum of 7 and 5 times a number n. 2

19. Find the area of a rectangle of length 25 m and breadth 12 m. 2

20. A solid has 6 faces, 8 vertices and 12 edges. Verify Euler's formula. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. State the number of faces, vertices and edges of a triangular pyramid and verify Euler's formula. 3

22. The marks of 6 students are 42, 43, 48, 80, 34, 23. Find the mean mark and the number of students who scored above the mean. 3

23. Evaluate $3p + 4q - 5$ for $p = 6$, $q = 2$. 3

24. Find the area of a parallelogram with base 14 cm and height 9 cm. What is the area of the triangle formed by half of it (cut along a diagonal)? 3

OR

In the Sierpinski triangle, how many shaded triangles are there at stage 2? How many more are there at stage 3? 3

25. Arrange the data 7, 55, 70, 44, 69, 21, 50, 37, 6 in ascending order and find its median. 3

OR

Simplify $9x + 4 - 5x + 8 - 2$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. A Koch curve begins with a straight line of length 9 cm (stage 0). At each stage every segment is replaced by 4 segments each one-third as long. Find the number of segments, the length of each segment and the total length at stage 2. 5

OR

For the data 6, 25, 25, 4, 23, 8, find the mean, median, mode and range. 5

27. Think of a number n. Multiply it by 3, add 5 and then subtract 5. Show the result algebraically. What is the result when $n = 4$? 5

OR

A square plot has a perimeter of 24 m. Find its side and area. How many square tiles of side 1 m are needed to cover it? 5

END OF QUESTION PAPER — **Total: 50 Marks**